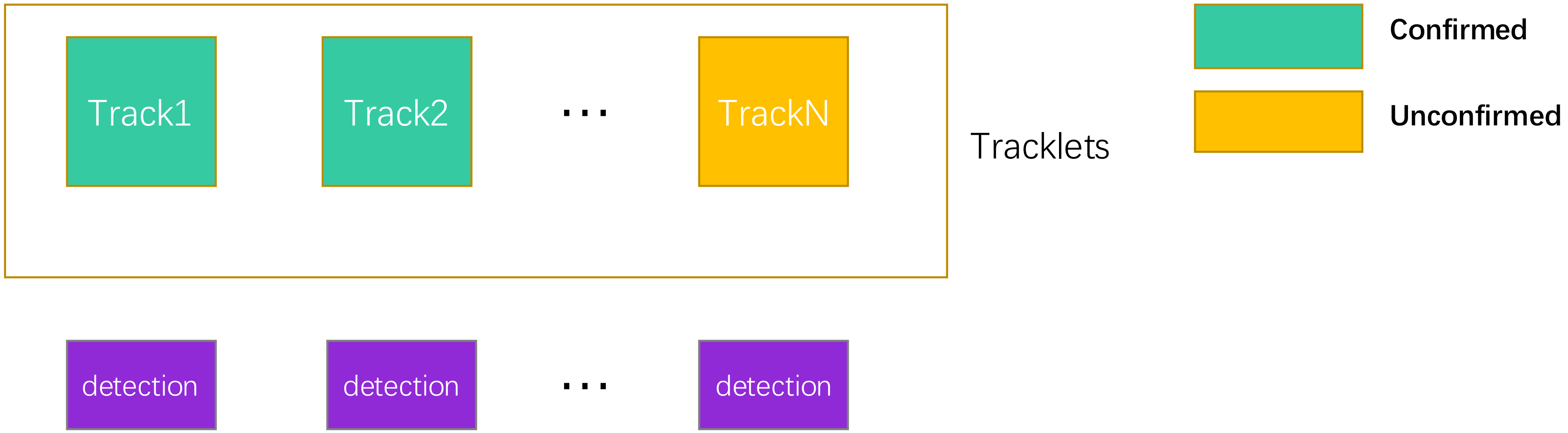
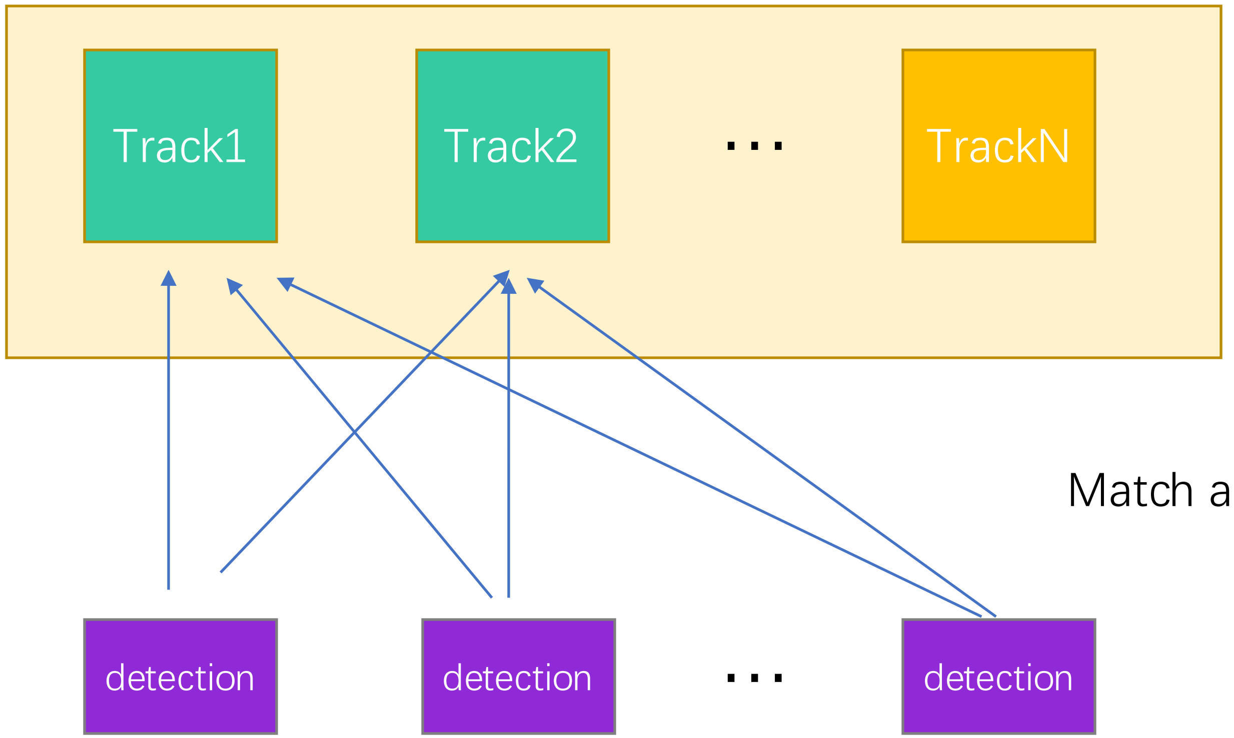
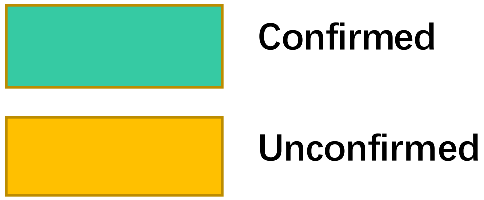
Multi-target Single-camera Tracking



Multi-target Single-camera Tracking

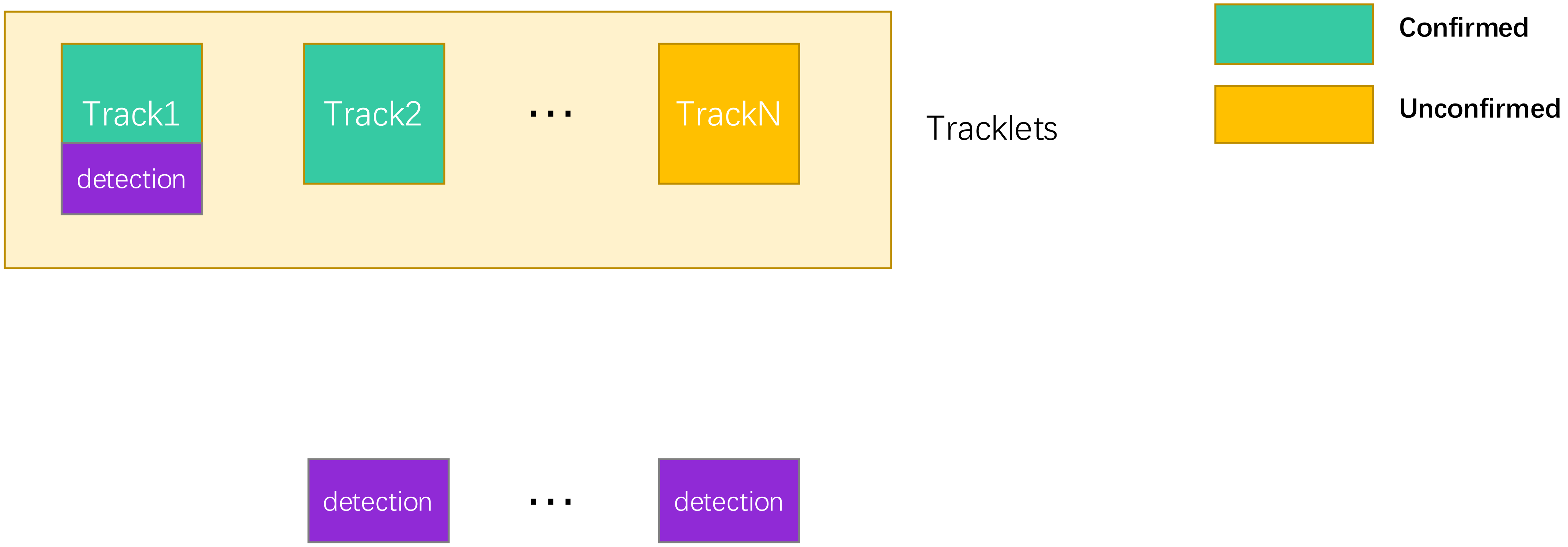


Tracklets

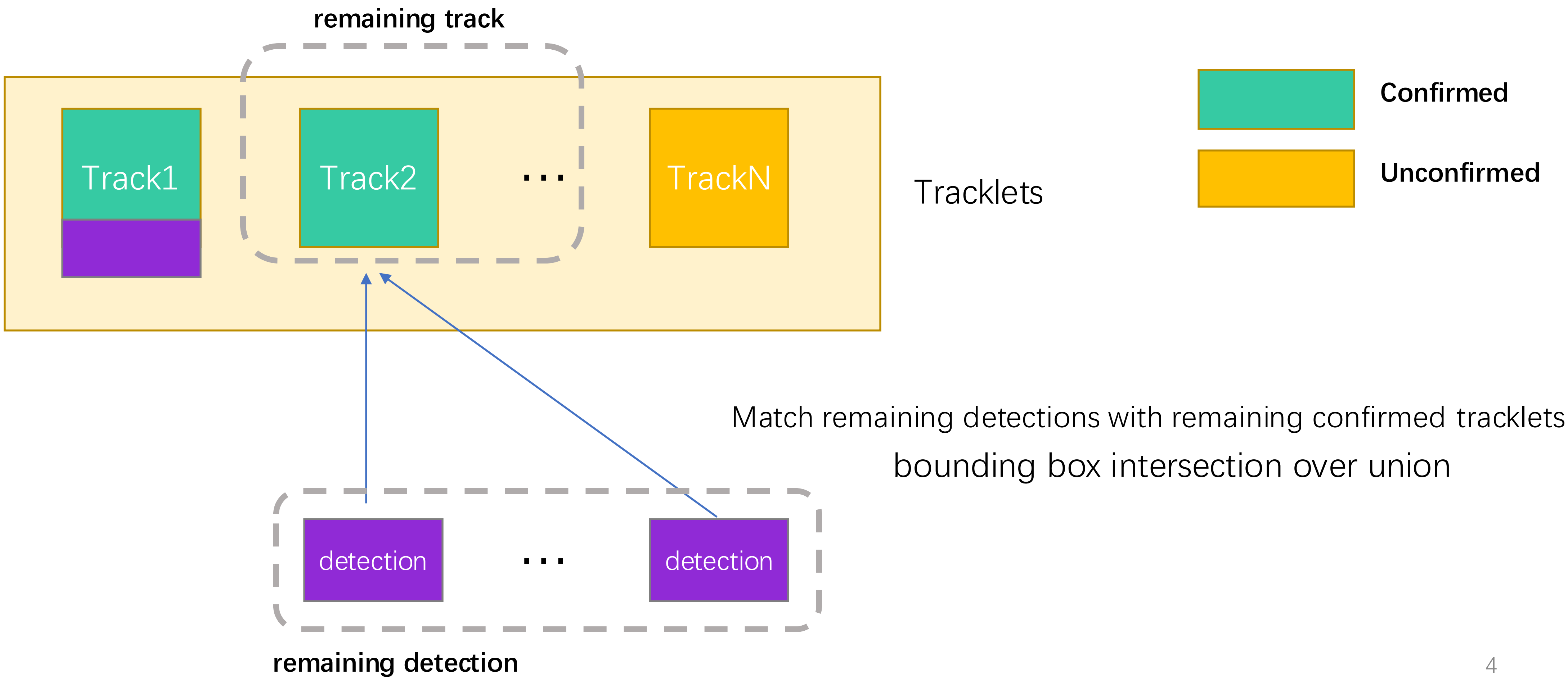


Match all detection with confirmed track
feature similarity

Multi-target Single-camera Tracking



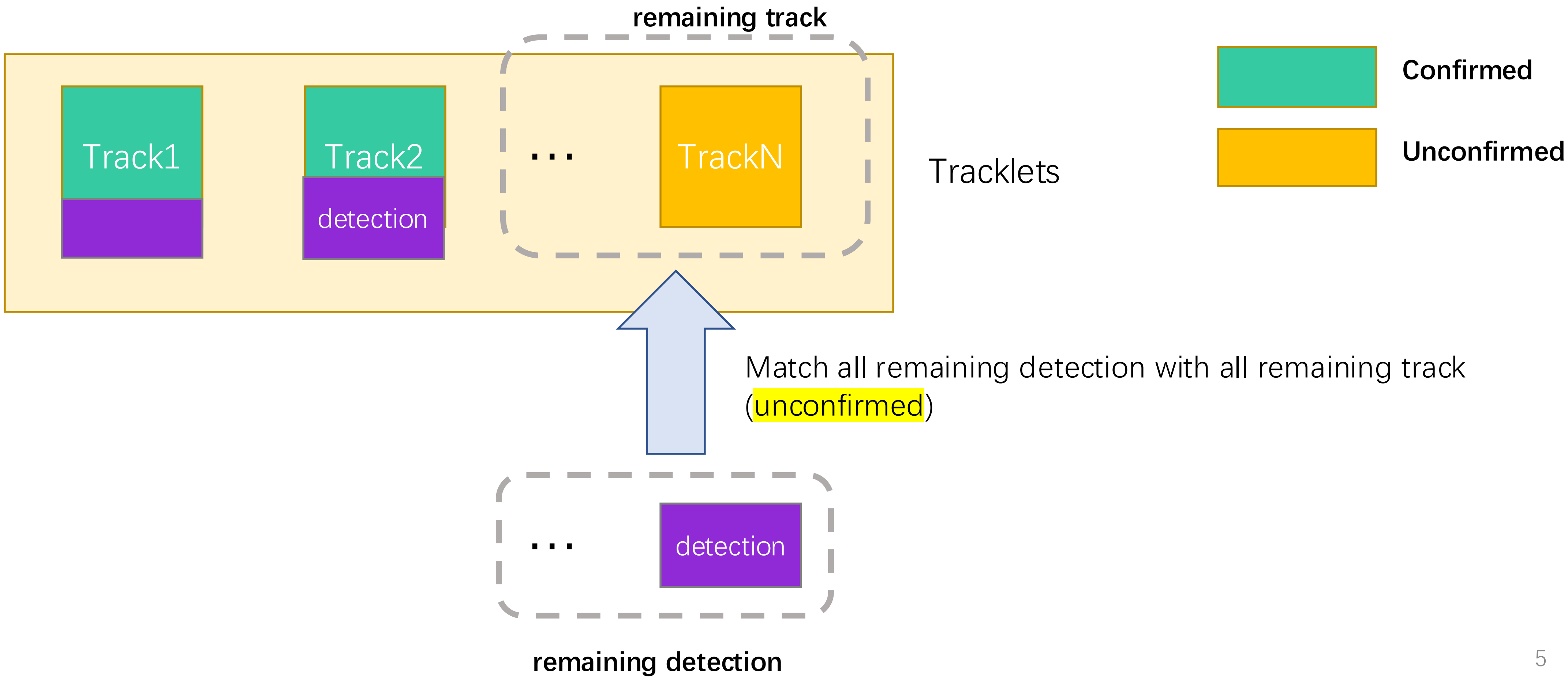
Multi-target Single-camera Tracking



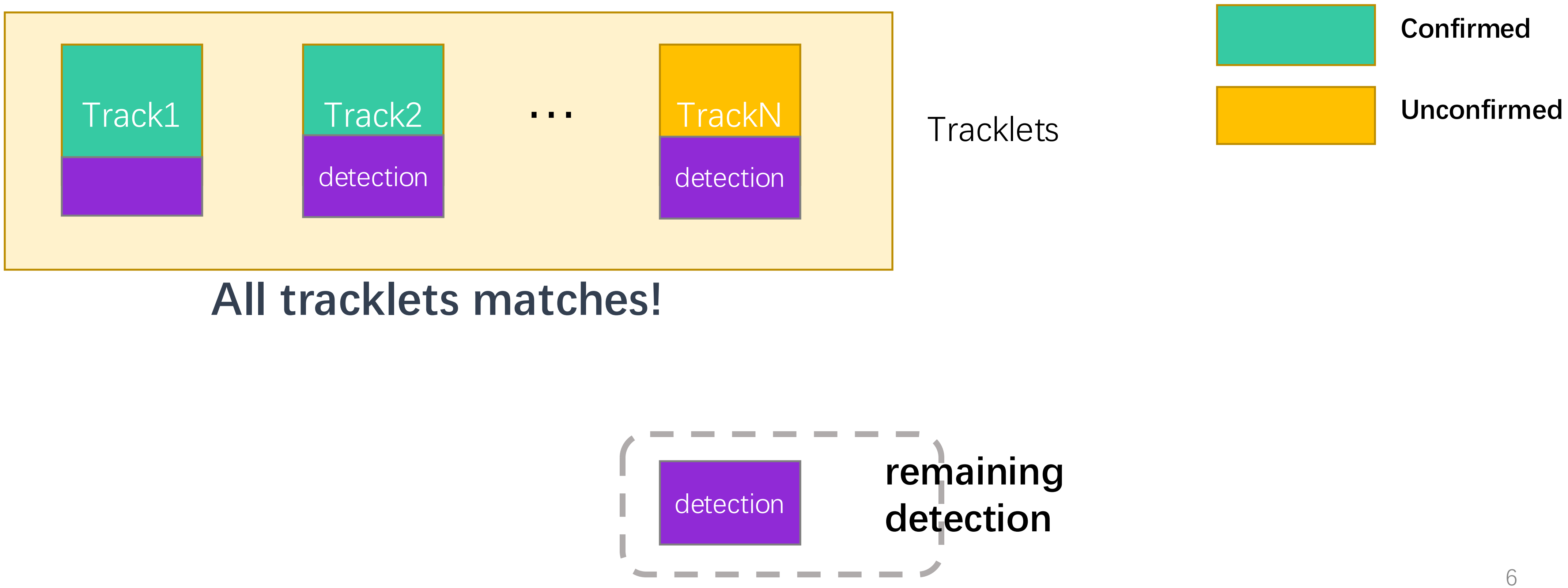
Match remaining detections with remaining confirmed tracklets

bounding box intersection over union

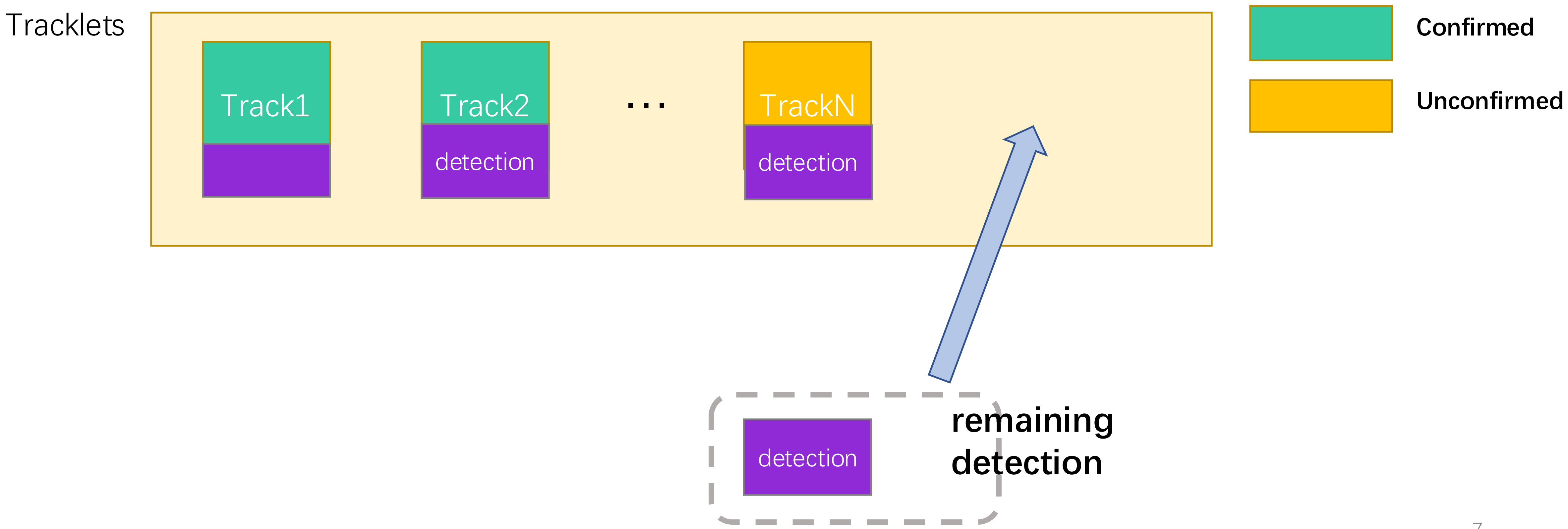
Multi-target Single-camera Tracking



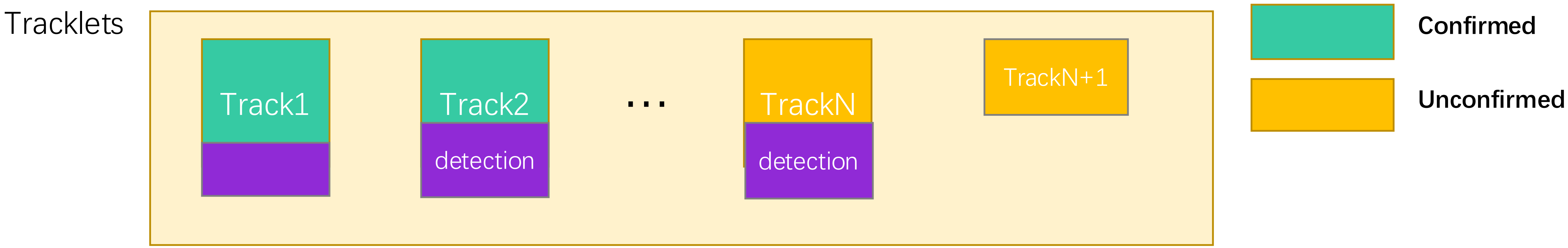
Multi-target Single-camera Tracking



Multi-target Single-camera Tracking

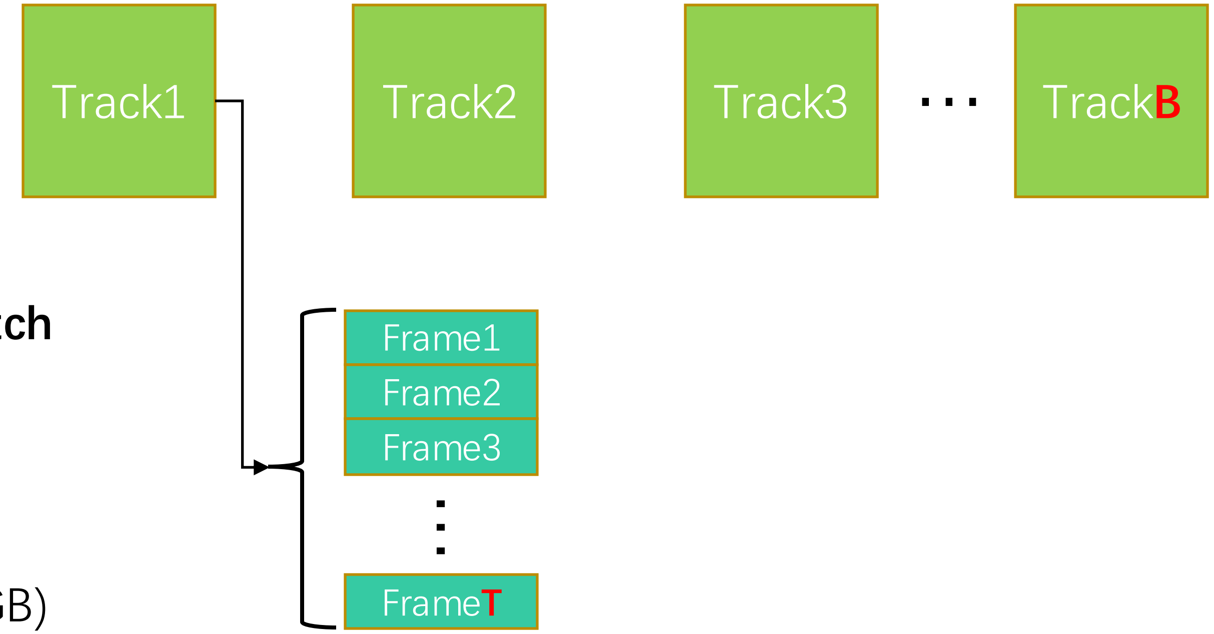


Multi-target Single-camera Tracking



Multi-target Single-camera Tracking Complete

Vehicle Re-identification



ReID training input is a Tracklet Batch
T × B × W × H × 3

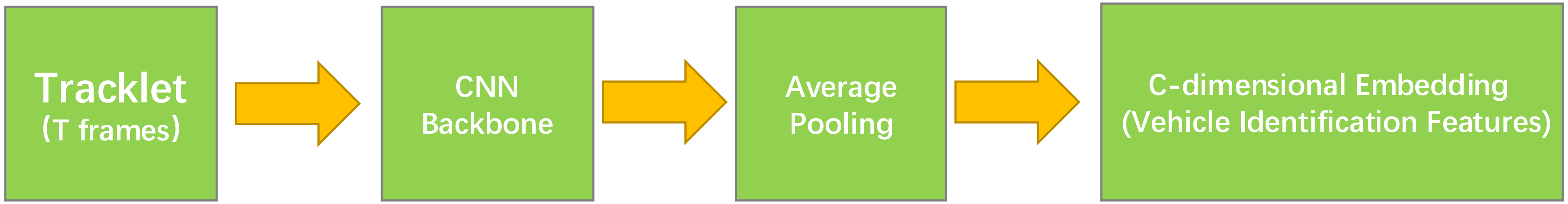
B: Number of tracks in a batch
T: Number of frames per track
W × H × 3: Size of a single image (RGB)

Vehicle Re-identification

Used CNN network N_feat to extract features F_{B×C}

B tracklets → Output B embeddings

Each embedding is a C-dimensional vector



Vehicle Re-identification

ReID is trained using Aggregation Loss.

$$loss = \alpha \times loss_{tr} + \beta \times loss_{xe}$$

Cross Entropy Loss (Classification Loss) Classifies different vehicles as different categories.

Triplet Loss (Metric Learning Loss)

Zooms in: Same vehicle

Zooms out: Different vehicles

$$loss_{tr} = \max(0, loss_{hp} + loss_{hn} + \lambda)$$

Hard Positive

Sample with the greatest exterior **differences** within the **same** vehicle.

$$loss_{hp} = \sum_{i=1}^{V} \sum_{j=1}^{B_i} (\max_{k=1}^{B_i}(D(N_{feat}(I_j^i), N_{feat}(I_k^i))))$$

Hard Negative

Sample with the **most similar** appearance among **different** vehicles.

$$loss_{hn} = -\sum_{i=1}^{V} \sum_{j=1}^{B_i} (\min_{\substack{p=1\ldots v \\ q=1\ldots B_p \\ p \neq i}} (D(N_{feat}(I_j^i), N_{feat}(I_q^p))))$$

Vehicle Re-identification

ReID is trained using Aggregation Loss. $loss = \alpha \times loss_{tr} + \beta \times loss_{xe}$

Cross Entropy Loss (Classification Loss) Classifies different vehicles as different categories.

Triplet Loss (Metric Learning Loss)

Zooms in: Same vehicle

Zooms out: Different vehicles

$$loss_{tr} = \max(0, loss_{hp} + loss_{hn} + \lambda)$$

Hard Positive

Sample with the greatest exterior **differences** within the **same** vehicle.

Hard Negative

Sample with the **most similar** appearance among **different** vehicles.

ReID Inference Phase

The reasoning process is divided into two parts:

- **Query Set** The tracklets for which the identity needs to be determined.
- **Gallery Set** A database of historical tracklets features.

Calculate the feature distance matrix between Query and Gallery

- Small distance → Same vehicle
- Large distance → Different vehicles

The system only performs matching under the following conditions

- Reasonable timing
- Adjacent camera topologies
- Successful matching → Merged into a single global ID

Failed matching → Treated as a new vehicle

Multi-target Cross-camera Tracking

Linking tracklets from different cameras to a unified Global ID

Inputs

- Tracklets obtained from single-camera tracking
- ReID output: Feature similarity (distance matrix)
- Camera topology
- Temporal constraint

Output: Consistent Global Vehicle ID across cameras

Global ID merging criteria

- ReID distance is sufficiently small
- Occurrence time is reasonable
- Cameras are physically adjacent or reachable